Response to Reviewers

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One resubmission. Each item: (a) concern, (b) response, (c) Overleaf action.
Numbers: `revision_gap_closure/09_values/MANUSCRIPT_VALUES.json` (23 Oct 2026).

We thank the Associate Editor and the six reviewers. Every numbered concern is answered with new evidence or an explicit, scoped limitation. Three artefacts replace the previous 31-binary / 3,100-launch / 10 s corpus experiment:

1. Functional oracles (guest, 22 Oct 2026): matched AntiDebug 3+3 (`04_oracles/from_vm/antidebug_n3.csv`); `-da` on all five oracles; Module32 / waits. Timing N=30 was not rerun (hash `76208218...`).

2. SkewMask timing oracle, fixed N=30, all attempts retained on TestOverhead.exe. GO median 3,083.210 ticks (11/30 below 3,000); GO_DO 2,104.770 (30/30); TMW0 2,433.285 (30/30).

3. Corpus campaign: 100 PE □ 9 □ 10 = 9,000/9,000 JSON, seed 705851366, Pin 3.28.98749, snapshot `tomware_go`. Ladder NAT / EMPTY / TMW0 / five modules / TMW_ALL. Windows: 120 s short-lived, 60 s persistent. Official tables: `09_values/MANUSCRIPT_VALUES.json` (audited), not the un-audited `processed/` folder alone.

Overleaf must replace every 31 / 3,100 / 10 s / 155/155 / ?87.2% cell. The 6,517 collector labels are exit code unknown, not crashes.

Reviewer 1

R1.1 □ AntiDebug was not matched

(a) Reviewer concern. Baseline used ?gdb; treatment used ?dd without ?gdb.

(b) Response. Matched oracle, N=3: `-gdb -q` reported BeingDebugged=1, NtGlobalFlag=1, IsDebuggerPresent=1 in 3/3 runs; `-gdb -dd -q` reported all zero in 3/3. NAT and TMW0 without ?gdb are negative controls (all zero). Scope: those two PEB fields plus the oracle□s IsDebuggerPresent readout. Corpus ?dd runs are not proof that a sample queried the PEB.

(c) Action in Overleaf. Figure 2 / Table 4 = this matched pair. Do not label AntiDebug unvalidated.

R1.2 □ SkewMask timing selection bias

(a) Reviewer concern. Nine of 31 checks were re-executed until the score fell below the threshold.

(b) Response. Replaced by pre-fixed N=30. GO: 11/30 (36.7%), median 3,083.210 (2,722□3,845). GO_DO: 30/30, median 2,104.770 (2,073□2,146). TMW0: 30/30, median 2,433.285. The 9/31 figure must not appear. Corpus TMW_DO kept all 1,000 outcomes.

(c) Action in Overleaf. Abstract and Table 4: compensation rate 11/30 vs 30/30, not a binary □effective.□

Table 1. Timing oracle, fixed N=30, all attempts retained.

Arm	Knobs	N	Median ticks	Min□max	Below 3,000
TMW0	?q	30	2,433.285	2,168□2,750	30/30 (100%)
GO (baseline)	?go ?q	30	3,083.210	2,722□3,845	11/30 (36.7%)
GO_DO (treatment)	?go ?do ?q	30	2,104.770	2,073□2,146	30/30 (100%)

R1.3 □ No experimental benchmark against Arancino

(a) Reviewer concern. Without a direct comparison, modularity remains an assertion.

(b) Response. Architectural table only (Arancino, COBAI, Peekaboo). Public Arancino is Pin 2.14 + VS 2010 and 32-bit only. This campaign is Pin 3.28 and 100 PE x64. We do not claim superiority. Future work (after this resubmission): 32-bit oracles / optional 32-bit corpus on Pin 3.28 ia32, then a bake-off on a shared 32-bit protocol. The 9,000-run campaign is not that bake-off.

(c) Action in Overleaf. Keep Table 1 (related systems). □VIII: add the 32-bit / Arancino items; do not rank tools.

R1.4 □ Malicious corpus small and undocumented

(a) Reviewer concern. Fourteen unclassified samples.

(b) Response. RQ2 uses 100 PE (50+50; 25+25+25+25), all x64. `corpus.csv` lists SHA-256, source (Windows_Environment / MalwareBazaar, 2026-10-14), architecture, size, profile, targeted evasion (none / unverified). family_or_product is a coarse label (Console/GUI App; Trojan_Bazaar / Backdoor_Bazaar), not a VT family. We do not claim the malware implemented the five hooked checks (Etapa 5: ProcessEnum 0/269; memcmp 0 PIN_ matches).

(c) Action in Overleaf. Replace every 17+14 / 31 / 3,100 / 155 sentence. Condensed Table 3; point to corpus.csv.

R1.5 □ Medians without CI; ?87.2% working set

(a) Reviewer concern. Large SkewMask resource deltas may reflect altered execution, not a gain.

(b) Response. Median of per-binary medians, IQR, bootstrap 95% CI (2,000; seed 705851366); Wilson on rates. Under the 60 s window SkewMask increases wall-clock (benign persistent TMW0 2.2 s ? TMW_DO 60.8 s; paired +55,452 ms [30,685, 57,031]). Survival: benign 8.0% ? 52.8%; malicious TMW0 32.8% ? TMW_DO 57.6%. WS TMW_DO?TMW0: benign P ?3.2 MB [?58.3, +0.4] (includes 0); malicious P ?22.7 MB [?94.9, ?10.7] (n_pairs=23). CPU rises because the process stays scheduled. Not a performance gain. Delete ?87.2 / ?38.5.

(c) Action in Overleaf. Replace Table 7 with absolute paired deltas. Negative WS ? improvement.

R1.6 □ Narrow interception versus the title

(a) Reviewer concern. NtQuerySystemInformation and RDTSC bypass the modules.

(b) Response. Agreed. □Mitigating evasive techniques□ means selected indicators on monitored interfaces.

(c) Action in Overleaf. Keep the scoped □VIII. No transparency claim.

Reviewer 2

R2.1 □ Share the matched AntiDebug rerun

(a) Reviewer concern. Rerun with ?gdb in both arms.

(b) Response. Done (R1.1), 3+3, antidebug_n3.csv.

(c) Action in Overleaf. See R1.1.

R2.2 □ Fixed-N timing, retain all, CIs

(a) Reviewer concern. Predefine repetitions; do not drop misses.

(b) Response. Oracle N=30 all kept (Table 1). Corpus TMW_DO: 1,000 kept, bootstrap CIs.

(c) Action in Overleaf. Do not write □SkewMask is effective□ as a binary verdict.

R2.3 □ Native / empty / disabled-module baselines

(a) Reviewer concern. Incremental-only baselines hide Pin□s cost versus native.

(b) Response. Nine arms including NAT, EMPTY, TMW0. The 9,000-run campaign omitted bare Pin without ?t because that day□s probe failed (`pin_condition_valid=false`). An isolated probe on 22 Oct completed with exit 0 □ we do not claim Pin-without-tool is always impossible. Benign short-lived median-of-medians wall-clock: NAT 0.26 s, EMPTY 1.48 s, TMW0 1.97 s, TMW_DP 5.53 s, TMW_DO 35.0 s, TMW_ALL 36.7 s.

(c) Action in Overleaf. Baseline-ladder table in □VI. Strike future-work item 5.

R2.4 □ Auditable corpus table

(a) Reviewer concern. SHA-256, source, family, packer, architecture, profile, evasion.

(b) Response. Those columns exist. Family is coarse; malware targeted_evasion = unverified. No invented VT names.

(c) Action in Overleaf. Condensed table in the paper.

R2.5 □ Arancino / COBAI / Peekaboo

(a) Reviewer concern. State what is unique.

(b) Response. Architectural table (R1.3). Evidenced distinction: independent knobs + nine-rung ladder + ?da on one protocol.

(c) Action in Overleaf. No experimental ranking. Future work: shared 32-bit protocol.

R2.6 □ Behavioral fidelity

(a) Reviewer concern. Add API traces, coverage, process activity.

(b) Response. The 9,000 JSON have counters off. A later 120-run external observer (exit, stdout hash, children, new files; no FidelityProbe) found TMW0 vs TMW_ALL same exit+stdout in 22/30 pairs. This is not API or BBL equivalence.

(c) Action in Overleaf. Compatibility stays operational. Keep fidelity as a limitation / future work item 4.

R2.7 □ Compatibility too weak

(a) Reviewer concern. Completion or 10 s alive ? behavioural equivalence.

(b) Response. 155/155 @ 10 s is withdrawn. Persistent criterion = 60 s controlled_termination. Survival n=250, Wilson 95%: benign NAT 8.0% [5.2, 12.0], TMW_DO 52.8% [46.6, 58.9], TMW_ALL 56.4% [50.2, 62.4]; malicious NAT 54.4% [48.2, 60.5], EMPTY 32.4% [26.9, 38.4], TMW0 32.8%, TMW_DO 57.6%, TMW_ALL 64.0% [57.9, 69.7]. Short-lived protocol termination ?96.8% in every arm. The 6,517/9,000 dominant short-lived label is exit code unknown (collector bug), not a tool crash. Launch failures 251/9,000.

(c) Action in Overleaf. Delete 155/155. Table 5 = outcome classes. Abstract must not say all combinations passed.

R2 extra □ combined mode; five checks; novelty

(a) Reviewer concern. No ?da; no proof malware used the five techniques.

(b) Response. TMW_ALL is in the 9,000-run tables and tracks TMW_DO on persistents (SkewMask dominates). Etapa 5 (different DLL, 30 PE): AntiDebug surface 269/269 is load-time PEB sanitize; ProcessEnum 0/269; memcmp 146 intercepts and 0 PIN_ matches. RQ2 stays operational.

(c) Action in Overleaf. Add a TMW_ALL row. One sentence: the 100-sample table is not evidence each binary exercised the check.

Reviewer 3 (minor)

R3.1 □ Matched AntiDebug

(a) Reviewer concern. Match or label unvalidated.

(b) Response. Matched 3+3 (R1.1).

(c) Action in Overleaf. Do not label unvalidated.

R3.2 □ 29% re-execution

(a) Reviewer concern. Explain the 29% (9/31).

(b) Response. Old protocol only. Current: N=30, zero re-executions, 11/30 vs 30/30.

(c) Action in Overleaf. Delete 29% from the abstract.

R3.3 □ Uncertainty

(a) Reviewer concern. Add CIs or call results descriptive.

(b) Response. Wilson + bootstrap. No p-value hunt.

(c) Action in Overleaf. Label tables □descriptive with interval estimates.□

R3.4 □ Related-systems table

(a) Reviewer concern. Concise comparison table.

(b) Response. Already in the revision PDF (Table 1).

(c) Action in Overleaf. Keep it.

R3.5 □ Language

(a) Reviewer concern. Typos / spacing.

(b) Response. Noted.

(c) Action in Overleaf. Decimal points (not locale commas). Fix Table 8 wrap.

Reviewer 4

R4.1 □ Matched AntiDebug now, not future work

(a) Reviewer concern. Do the oracle.

(b) Response. Done (R1.1).

(c) Action in Overleaf. Remove from future work.

R4.2 □ All timing outcomes; rate in abstract

(a) Reviewer concern. Fix N; report every outcome.

(b) Response. Table 1; abstract 11/30 vs 30/30.

(c) Action in Overleaf. No single □below threshold□ verdict.

R4.3 □ ?87.2% / ?38.5%

(a) Reviewer concern. Largest effect may be a near-halt.

(b) Response. The 60 s campaign is that test (R1.5). We adopt Reviewer 4□s reading: □still alive□ is not compatibility.

(c) Action in Overleaf. Delete ?87.2 / ?38.5.

R4.4 □ Prospective metric-validity rule

(a) Reviewer concern. Wall-clock accepted for some modules and N/A for others, with no reconstructable rule.

(b) Response. What was in experiment.config.json before the 9,000 runs: short-lived timeout 120 s; persistent window 60 s with controlled termination; 10 repetitions; failures/timeouts retained; no outcome-selected rerun; restore per block. What was not in the

config: the resource filter `sample_count>0` (applied post hoc in the audit). We say so. ProcessEnum wall-clock is cost (benign SL TMW_DP?TMW0 +3,844 ms). Launch failures (251) are recorded outcomes.
(c) Action in Overleaf. Paste this distinction into □VI. Do not claim sample_count>0 was a priori.

R4 extra □ native, figures, hooks, ?da

(a) Reviewer concern. No native baseline; no hook counts; no ?da.
(b) Response. NAT/EMPTY/TMW0/?da are in the 9,000-run ladder. Hook counters were off in those JSON. A 600-run diagnostic (other DLL) is reported as such and is not mixed with the 9,000.
(c) Action in Overleaf. Normalize figure decimals.

Reviewer 5

R5.1 □ Debugger configurations

(a) Reviewer concern. Match debugger configs.
(b) Response. Matched (R1.1).
(c) Action in Overleaf. See R1.1.

R5.2 □ Moderate behavioural-preservation claims

(a) Reviewer concern. Compatibility ? equivalence of syscalls / CFG / effects.
(b) Response. Agreed. Etapa 6: 22/30 pairs same exit+stdout under TMW0 vs TMW_ALL. The 6,517/9,000 label is unknown exit, not crash.
(c) Action in Overleaf. Abstract and □VI-D: operational criteria ? behavioural equivalence.

R5.3 □ Performance may reflect state change

(a) Reviewer concern. Say results are descriptive.
(b) Response. Done (R1.5). SkewMask wall-clock is the example.
(c) Action in Overleaf. See Table 7 replacement.

R5.4 □ References

(a) Reviewer concern. Add recent DBI-evasion papers.
(b) Response. Optional. IEEE Access does not require reviewer-suggested cites.
(c) Action in Overleaf. Add only papers the authors have read.

R5.5 □ Scope of conclusions

(a) Reviewer concern. Stay within oracles and corpus.
(b) Response. Five oracles + 100 PE / nine conditions. No transparency, no Arancino win, no □malware used these five checks.□
(c) Action in Overleaf. Keep that sentence in □VIII.

Reviewer 6

R6.1 □ AntiDebug configs differ

(a) Reviewer concern. Hard to interpret.
(b) Response. Matched (R1.1).
(c) Action in Overleaf. See R1.1.

R6.2 □ Timing re-execution

(a) Reviewer concern. Selection bias.
(b) Response. Fixed-N=30 (Table 1).
(c) Action in Overleaf. See R1.2.

R6.3 □ Weak operational compatibility

(a) Reviewer concern. Criterion ? equivalence or transparency.
(b) Response. 155/155 removed; 60 s survival and exit-unknown mix reported (R2.7).
(c) Action in Overleaf. No behavioural-equivalence claim.

R6.4 □ No native baseline; only medians

(a) Reviewer concern. Need native + CIs.

(b) Response. NAT + EMPTY + Wilson/bootstrap (R2.3, R3.3).

(c) Action in Overleaf. New ladder and interval tables.

R6.5 □ Novelty versus existing frameworks

(a) Reviewer concern. Novelty should be discussed and tested.

(b) Response. Architectural table; no bake-off (R1.3). Novelty = modular evaluation + protocol. 32-bit/Arancino experiments after this resubmission.

(c) Action in Overleaf. Do not claim an experimental ranking.

Corpus tables for Overleaf (do not invent other cells)

Table 2. Persistent window-survival, n=250, Wilson 95% CI.

Group	NAT	EMPTY	TMW0	TMW_DO	TMW_ALL
Benign persistent	8.0% [5.2, 12.0]	11.2% [7.9, 15.7]	9.6% [6.5, 13.9]	52.8% [46.6, 58.9]	56.4% [50.2, 62.4]
Malicious persistent	54.4% [48.2, 60.5]	32.4% [26.9, 38.4]	32.8% [27.3, 38.8]	57.6% [51.4, 63.6]	64.0% [57.9, 69.7]

Table 3. Median of per-binary median wall-clock (ms).

Stratum	NAT	EMPTY	TMW0	TMW_DP	TMW_DO	TMW_ALL
Benign short-lived	261	1,485	1,974	5,534	34,990	36,695
Benign persistent	254	2,180	2,190	6,094	60,770	60,746
Malicious short-lived	6,165	10,332	14,004	13,747	87,009	91,086
Malicious persistent	60,669	16,869	15,304	18,891	60,670	60,802

Table 4. Paired treatment ? TMW0, bootstrap 95% CI.

Contrast	Stratum	? wall-clock (ms)	? peak WS (MB)
TMW_DP ? TMW0	Benign SL	+3,844 [3,427, 4,088]	+0.5 [0.1, 1.0]
TMW_DO ? TMW0	Benign SL	+31,669 [13,744, 51,072]	+0.7 [?0.2, 2.2]
TMW_DO ? TMW0	Benign P	+55,452 [30,685, 57,031]	?3.2 [?58.3, +0.4]
TMW_DO ? TMW0	Malicious P (n_pairs=23)	+37,387 [26,146, 42,770]	?22.7 [?94.9, ?10.7]
TMW_ALL ? TMW0	Benign P	+55,512 [38,005, 56,972]	?5.1 [?74.6, 0.0]
TMW_DM ? TMW0	Malicious SL	wall CI includes 0	+0.451 [0.049, 0.805]

9,000 outcomes: exit-unknown 6,517; window 1,355; timeout 877; launch_failed 251. True crash in the 9,000: 0.

Do not write in Overleaf

□ 155/155, 10 s window, 31 / 3,100 as the current protocol.
□ ?87.2% memory or ?38.5% CPU as a gain.
□ □SkewMask always hides Pin□ (baseline 11/30).
□ □The 100 malware samples used the five checks.□
□ □TOMWare.M outperforms Arancino.□
□ Behavioural equivalence, transparency, or □short-lived runs are mostly crash-exit.□
□ sample_count>0 was a priori in experiment.config.json.